



Documentation

Electronics
BlueVision NG Automation Suite
IOU Project Data

E532864/01E

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1.1 E532864/01E BlueVision NG Automation Suite - IOU Projektierungsdaten

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1 Introduction

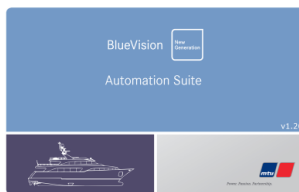
This user documentation describes handling and functioning of the “*IOU application engineering data generator and BDM feature*” of the BlueVision New Generation Automation Suite 1.24 (or above). This functionality requires at least one IOU within the system and allows IOU application engineering data to be changed, generated and made available to the BDM server integrated in the Automation Suite.

1.1 Note / Prerequisites

1.1.1 Notebook

Processor:	Dual Core / 2 GHz
Main memory:	2GB RAM
Screen:	1024x768 (minimum) / 1280x1024
Network:	Ethernet 100 Mbit/s
Operating system:	Microsoft Windows 7 or Windows 10

1.1.2 Blue Vision New Generation Automation Suite



The latest version of Automation Suite can be obtained via the Business Portal:

<https://services.mtu-online.com>

1.1.3 Ixxat CAN-to-USB / X00E50210754



Ixxat CAN hardware with VCI device drivers Version 3.5 or 4.0 is required for operating the BDM functionality of Automation Suite.

Download link as of the documentation's publication date:

<https://www.ixxat.com/de/support/downloads-dateien-dokumente/treiber/vci-v3-treiber>

<https://www.ixxat.com/de/support/downloads-dateien-dokumente/treiber/vci-v4-treiber>

1.1.4 Dialog cable / 5355306734



The dialog cable is required for connecting the Ixxat CAN hardware with the IOU.

2 Obtaining IOU application data / BDM data

If there is a connection to a plant via Automation Suite, the IOU BDM data is downloaded onto the PC –provided an IOU is configured for the plant.

If it is detected that these data have not yet been transferred to the IOUs when connecting to the installation, this is reported automatically.

3 Modification of IOU application data

The IOUs available in the system can be selected in the plant configurator and configured accordingly. If the IOU configuration is changed, new BDM data is generated and saved both on the PC and in the plant.

4 BDM download

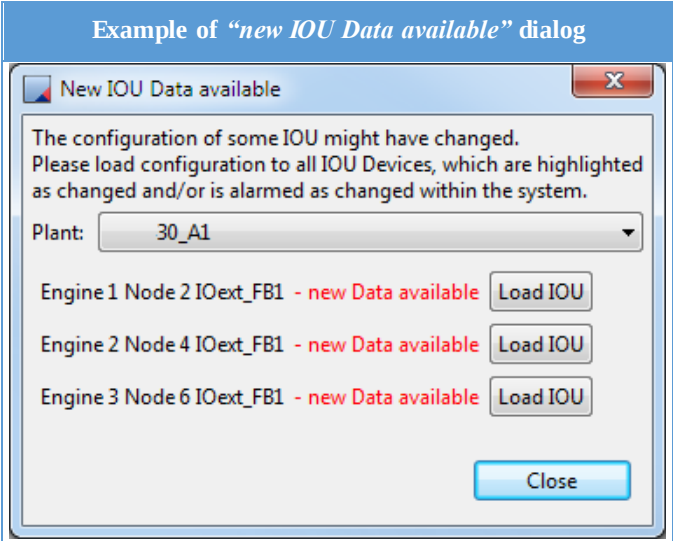
The BDM function can be started via the “Load IOU” menu item.

If there is an online connection at this time, the BDM data is selected automatically. In offline mode, the plant version for which BDM data shall be loaded needs to be selected.

The “Load IOU” function is available for every CAN bus with IOUs. Baud rate is automatically preconfigured and all data modules are loaded automatically.

Note:

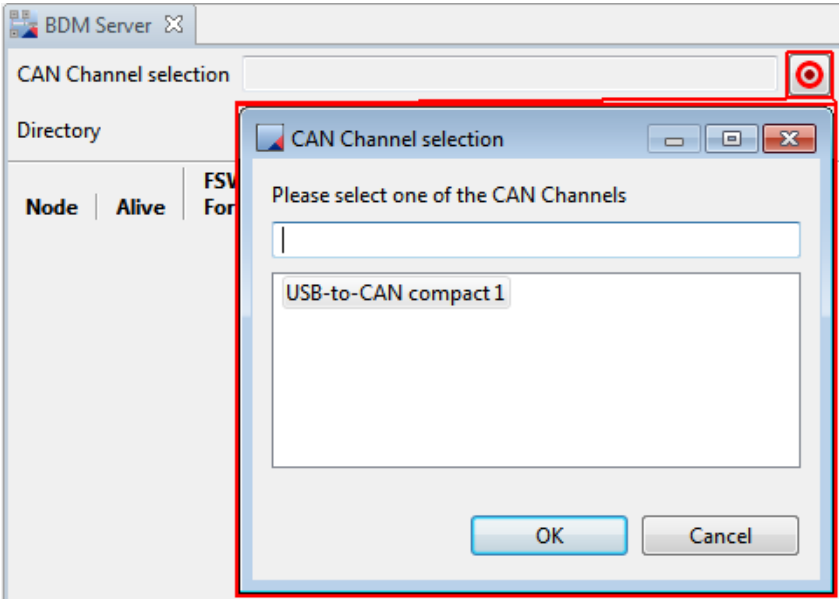
If BDM data of an IOU has not yet been loaded onto the installation it is highlighted with “new Data available”.



Note:
Node number and other settings of the device are described in the IOU device documentation.

4.1 Select channel

The  button can be used to select the IXXAT hardware CAN channel to be used:





4.2 Start BDM server

The **Start BDM Server** button starts the server .

4.3 Stop BDM Server

The **Stop BDM Server** button stops the server. Downloads which are in progress are aborted without any further notification.

The server is likewise stopped when the BDM Server View is closed.

4.4 View of download modules

After selecting a directory containing valid device data, the files available for download are shown based on node and type (FSW / PU):

Example - view of download modules

BDM Server

CAN Channel selectionUSB-to-CAN compact 1Baudrate125000Force All

Directory\\.\MTU\AutomationSuite\BDMServer\04.Work\A74\mpuStart BDM ServerStop BDM Server

Node	Alive	FSW Force	File	Status	Errors	PU Data Force	File	Status	Errors
2		☐	<no File>	-	0	☐	eaconf2.dat	Server offline	0
4		☐	sys_sw4.sys	Server offline	0	☐	eaconf4.dat	Server offline	0
5		☐	sys_sw_sam.sys	Server offline	0	☐	eaconf5.dat	Server offline	0
6		☐	sys_sw_mpu27.sys	Server offline	0	☐	eaconf6.dat	Server offline	0

1. 2. 3. 4. 5. 6. 7. 8. 9. 10.

TIM-ID: 0000127118 - 004



Meaning of the various columns:

Column	Module	Name	Meaning
1.	FSW/PU	Node	Number of the node for which data is available
2.	FSW/PU	Alive	Indicates whether a message was received from the node within a timeout (1200 ms)
3.	FSW	Force	Checkbox enabling forced downloading of the FSW data A check mark is automatically reset after a successful download
4.	FSW	File	Name of the FSW module
5.	FSW	Status	Status of the node, checksum or download progress (in regard of FSW data)
6.	FSW	Errors	Unsuccessful download counter (in regard of FSW data)
7.	PU	Force	Checkbox enabling forced downloading of the PU application engineering data A check mark is automatically reset after a successful download
8.	PU	File	Name of the PU data module
9.	PU	Status	Status of the node, checksum or download progress (in regard of PU data)
10.	PU	Errors	Unsuccessful download counter (in regard of PU data)

4.5 Force All

The **Force All** button makes it easier to set all check marks in the **Force** columns in the case of FSW and PU data. The checkboxes next to the respective download modules can be used for clearing. Selecting the same or a different directory as a source of device data resets all check marks.

4.6 Status of download modules

After starting the BDM server using the corresponding button, all download modules initially change to **Node not on bus** until either an incoming message is registered or the server is stopped again.

Subsequent status depends on the nature of the incoming message:

Status	Description
Node not on bus	Node is not active on bus
Checksum not verified	Device has been detected. Modules have yet to be checked to ensure up-to-dateness
Waiting for Download	Module was queued up for downloading
<Progress bar>	Download ongoing
Download finished	Module download completed. Up-to-dateness check pending
Checksum OK	The module on the device matches the application engineering data
Timed out	An error has occurred during download (value in Error column +1) Status changes automatically to “Node not on bus”